

RAY: An Interactive AI Art Experience Re-interprets Rayograph

Weidi Zhang

School of Design, Media and Immersive eXperience Center, Arizona State University

United States of America

zhangweidilydia@gmail.com

Abstract

This paper presents the conceptual and artistic background, intelligent system design, visualization strategies, and exhibition design of an interactive intelligent art experience: *RAY*. This artwork provides a responsive art experience that re-interprets Rayograph (photogram) – a 20th Century cameraless image-making technique – in the perspective of Artificial Intelligent (AI) surveillance to discuss the changing ontology of images. The intelligent interactive system translates live streaming video of audiences into an ever-evolving semantic Rayograph, revealing the hidden information in real-time. *RAY* bridges intelligent visualization with cameraless photography Rayograph to engage audiences with an interactive poetic experience that constructs the tension between humans and machines and embraces abstraction for meaning-making.

CCS Concepts

• **Computing methodologies** → Computer graphics.

Keywords

Experimental Visualization, Intelligent System Design, Interactive Art

ACM Reference Format:

Weidi Zhang. 2024. *RAY: An Interactive AI Art Experience Re-interprets Rayograph*. In *SIGGRAPH Asia 2024 Art Papers (SA Art Papers '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3680530.3695453>

1 Introduction

RAY is an intelligent, responsive system that converts the live-streaming image of audiences into a responsive visualization of new Rayographs (photogram) through learning over 3000 image pairs of human portraits and Rayographs. Similarly to the Rayograph objects that were exposed under the light to produce a shadow, which revealed hidden characteristics which helped create an image, we live in an age of ubiquitous cameras, where the intelligent systems hidden behind the cameras are used to evaluate the information of our bodies and life moments and make judgments based on the information they gather. *RAY* aims to provide a conceptual response to the changing notion of images and cameraless image-making technique under the gazes of machines through

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SA Art Papers '24, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1133-6/24/12

<https://doi.org/10.1145/3680530.3695453>

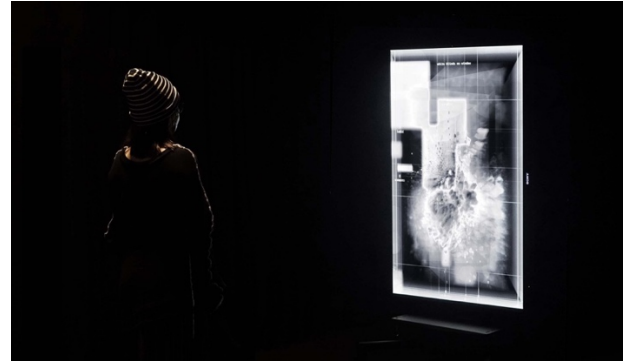


Figure 1: Installation View of RAY, 2022. (©Weidi Zhang. Photo: Weidi Zhang.)

re-interpreting Rayograph (photogram). *RAY* seeks to provide conceptual responses to the following questions: what is the shadow of our human bodies under the gazes of machines? What is the contemporary Rayograph? Who will be the creator(s)?

In this art installation (Figure 1), the intelligent system *RAY* constantly observes participants via a camera and transforms the live streaming data into a novel semantic Rayograph that evolves in real time. The intelligent system *RAY* consists of two parts: 1. generic image-to-image translation using conditional adversarial networks (GANs in the conditional setting). A neural network is trained to translate features from human portraits to a database of collected photograms of objects retrieved at Flickr API. 2. A computer vision system is used to localize and describe salient regions in images in natural language. Descriptive sentences are generated in real-time to describe the participants captured by the scanning camera in the installation. Those two outputs are visualized algorithmically in the stylistic aesthetics of 20th Century light painting and Rayographs.

2 Conceptual Background: Cameraless Photography and Automation

In 1922, Man Ray, a significant contributor to Dada and Surrealism movements, invented the term “Rayograph” to describe his cameraless darkroom-created photo images. He revealed the dreamlike characteristics of daily objects by placing them in photosensitized paper and exposing them to light. His Rayographs are situated in-between visual representation and abstraction, pointing to a new image-making method. Man Ray wrote in 1921 that he wanted to “make my photography automatic —to use my camera as I would a typewriter.” [MoMA Collection, 2019]

One hundred years later, Man Ray’s idea of automation with image-making is no longer a wild dream as the “overwhelming

majority of images are made by one machine to other machines, with humans rarely in the loop.” [Paglen, 2019] Countless images are no longer visual representations of the world made for humans to perceive but programmable data-based visualizations resulting from automatic operations. [Hoelzl and Marie, 2015] The notion of the cameraless image-making process shifted from Rayograph to the “rule-based activities tied to low-level cognitive procedures” [Cohen, 1979] and “plays a vital role in synchronic data-to-data relationships” [Hoelzl and Marie, 2015]. When an image is digitized, each element can be mapped, evaluated, compared, and linked to others, which provides networked collections that make metaphors. Digital images are programmable, operative, and related to system-making, as a result, it transforms images into constructed artifacts, whose representation reflects how culture has evolved in understanding images over time.

The autonomous process for image-making can be understood as a form of assemblage that contains imaging devices, the information/data they produce, and the infrastructure through which they operationalize [Mackenzie and Munster, 2019]. RAY takes inspiration from imaging devices hidden in public spaces, where intelligent machines evaluate information, make judgments, and captures life moments. “Instead of simply representing things in the world, machines and their images began to ‘do’ things in the world.” [Paglen, 2019] RAY creates a metaphor for the ubiquitous gazes of machines by asking: What is a contemporary Rayograph under the gazes of machines? What is the internal structure of a system that creates a new Rayograph? What are the objects for making the contemporary Rayograph, and who is the artist? Who are these systems designed to target? Are these systems capable of generating new aesthetics or cultural reproductions?

Taking these conceptual questions into consideration, I designed a multimodal intelligent system linking objects for photograms with human bodies under surveillance cameras as metaphors for the power of machines’ gazes. The system automatically transforms human portraits into new Rayographs with semantic meanings by training machine learning models to learn thousands of pairs of portraits and Rayographs. In addition, a computer vision system is used to generate semantic information to add a layer of meaning to the abstract evolving visuals. RAY reimagines 20th-century alternative photography through the lens of artificial intelligence and surveillance. It raises the critical question regarding the power behind technological systems and the objectification of humans. RAY aims to address the changing ontology of images under the gazes of machines by creating an intelligent system that incorporates automation with originality to generate a visualization of a human-machine reality (Figure 2).

3 Relevant Work

The image-to-image translation with conditional adversarial networks (Pix2Pix) has been demonstrated to be effective at synthesizing images from label mapping, colorizing images, and reconstructing objects from edge maps, among other tasks [Isola et al., 2017]. In the media art field, artists also implement the technique of image-to-image translation for creative practices.

New media artist Memo Akten’s series of interactive works – *Learning to See* – “use state-of-the-art machine learning algorithms



Figure 2: The intelligent system RAY generates the visualization (Left) in real-time through observing the live streaming video (Right) of a participant, 2022. (©Weidi Zhang. Photo: Weidi Zhang.)

to reflect on ourselves and how we make sense of the world” [Akten et al., 2019]. In the art installation, Memo uses a live camera to capture everyday objects and uses a number of neural networks to analyze the live streaming feed. The audience is invited to move the objects under the camera and see the visualization evolve based on the changing composition of the moving objects. The visualization changes every 30 seconds to different networks trained on different datasets: water, air, earth, and cosmos. Memo considers the way of seeing in the conscious mind as a reconstruction based on expectations and beliefs instead of a mirror image of the outside world [Akten et al., 2019].

Similar to *Learning to See*, RAY also used imaging devices, screen-based visualization, and intelligent systems to contribute automation and interaction. These two artworks differ in their artistic perspective and visualization development: 1. RAY addresses the way of seeing from a relatively different perspective. *Learning to See* focuses on how seeing relates to prior beliefs and knowledge. It provides the visualizations generated by different networks trained with varying datasets. On the other hand, RAY particularly focused on the poetic relationship between human bodies and objects for Rayograph, with a specific emphasis on alternative photographic processes (a modern cameraless image-making technique – Rayograph) related to image-making and camera culture. RAY criticizes the gaze of machines behind the surveillance camera and emphasizes the tension between revealing and concealing in the human-machine reality. 2. In *Learning to See*, the output by machines is presented as the final visual delivery on screens, while RAY further developed the machine’s visual output with the artist’s aesthetic decisions through an algorithmic process. RAY also incorporates a computer vision system to output language that makes meanings and metaphors for user interactions. RAY emphasizes a multilayered poetic assemblage created by the artist, the machine, and the participants, which combines automation with artistic originality.

4 Design Methodology

To achieve the goal of transforming the live streaming video of participants into an artistic visualization that conveys meanings,

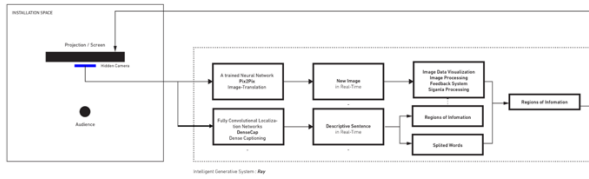


Figure 3: A system map describes the methodology of RAY, 2022. (©Weidi Zhang.)



Figure 4: Monochrome portraits of human faces (Left) with a gaussian blur, 2021. (©Weidi Zhang.)

the methodology of RAY mainly consists of three components (Figure 3): 1. Interactive intelligent system design, 2. Experimental visualization, 3. Art installation and sound design. The following subsections discuss these three components in detail, from the conceptual and technical development to the final presentation.

4.1 System design: an intelligent interactive system

The system of RAY utilized two neural networks to visualize the live streaming video of a participant captured by a camera set in the art installation: 1. generic image-to-image translation (pix2pix) using conditional adversarial networks [Isola et al., 2017]. 2. A computer vision system (DenseCap) is used to localize and describe salient regions in images through natural language [Johnson et al., 2016]. The system of RAY further processes the information output by neural networks, which is discussed in the following subsection of experimental visualization.

The first part of the system employs the technique of generic image-to-image translation with Conditional Adversarial Nets (Pix2Pix) [Isola et al., 2017]. A neural network is trained to convert human portraits into photograms. Then the learned model can generate new visualization from live-streaming images of audiences. The training dataset includes over 7000 photos of human portraits and photograms of daily objects retrieved from Flickr API. In order to experiment with visual results by embracing indeterminacy, the human portraits are intentionally turned monochrome with a gaussian blur (Figure 4) and randomly selected and paired with the corresponding images of photograms. After successfully training the neural network, the model is able to convert portraits into abstract Rayographs with interesting visual details (Figure 5).

Besides generating experimental visualizations converted from human portraits, the RAY system also produces localized descriptive sentences to describe audiences and their surroundings in the gallery space, attaching meanings to the new visualizations.

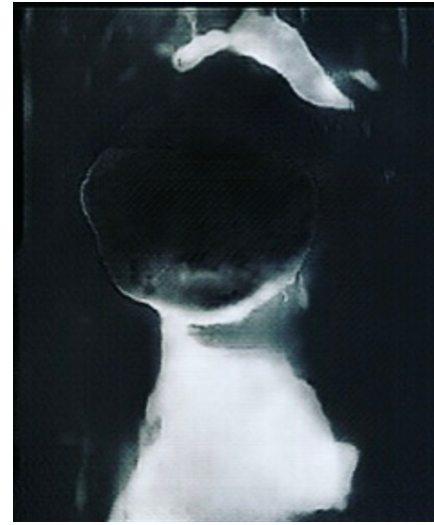


Figure 5: A generative image translated from an image of a human face using trained neural network RAY, 2021. (©Weidi Zhang.)

Inspired by the cut-up technique, a modern textual assemblage technique was first introduced by Brion Gysin and popularized by William Burrough's novel *Naked Lunch*. In *Naked Lunch*, Burrough randomly cut the words and sentences from different book pages, leading to the indeterminate yet the fascinating combination of the words from the following page with cut-out pages. The system implements a pre-trained fully convolutional localization network using dense captioning (DenseCap) [Johnson et al., 2016]. It requires a computer vision system to localize and describe the salient regions in images in natural language [Johnson et al., 2016]. Instead of directly presenting the corresponding descriptive sentences in the regions of the images, RAY uses the live streaming images data captured by a camera to determine which of the words in the generated sentences to present in the visualization. Incorporating partial randomness into the textual design creates a non-linear narrative that makes unpredictable metaphors for abstract visualization.

4.2 From alternative photography to experimental visualization

RAY emphasizes an assemblage of collective voices in an author-machines-audiences manner that merges artistic originality with automation to create a contemporary Rayograph. Thus, RAY does not directly present the image output of the trained neural network as the final deliverables. The system implements visualization strategies to develop the image data further through algorithmic computation with conceptual and aesthetic analysis.

A Rayograph is an image created by exposing light-sensitive photographic paper to light, where objects are placed on it to block areas of the paper. [McNeill, 2020] Variations of this technique are also implemented for scientific purposes such as X-Rays. Today, image-making practices are beyond merely taking pictures that record light, such as operative images and generative images that emphasize instructional practices and autonomous systems. The



Figure 6: A visualization generated by RAY based on observation of a participant via a camera, 2022. (©Weidi Zhang.)

artistic goal of visualization is to associate generative visualization with the modern cameraless image-making process by emphasizing the action of light through the actions of revealing and concealing information. Thus, RAY visualizes the machine-generated image data algorithmically within the tradition of light painting – a technique of taking a long exposure to a moving light source – to create the moving composition.

The RAY system instances each pixel of the neural network-generated images as a moving sphere. Image data of the positions (XY coordinates) and color (RGB values) determine the spheres' color and XY position. RAY implements a normalized sparse noise to trigger the spheres' movements in the Z-axis, which generates traces of actions. The red channel of image data multiplies the layers of noise (such as harmonic summation and Perlin noise) and determines the size and transparency of each moving sphere. Because the color information of each pixel is ever-changing, it turns each pixel into a moving brush stroke with unpredictable tonality, transparency, color, and size. The varying brushstrokes illustrate a sophisticated pattern which is later processed with layers of feedback systems and image processing techniques (such as alpha compositing and image difference) for fake motion blur with painterly lighting effects to create a poetic visualization (Figure 6).

In the visualization, participants are not expected to recognize what objects (such as hair clips, scissors, and others) are generated by machines. Rather, it aims to create a new Rayograph that uses pixel information from the generative image to create a new universal abstract language. RAY's abstract visualization dynamically moves and traces the machine's understanding of human bodies (Figure 7). This visual result is reminiscent of Man Ray's photographic work during the 1930s - *Space Writing* (1935) - in which Man Ray used a long exposure to capture a series of moments in a single image to illustrate the passage of time in the movement of his pen light. This is one of his works that creates dynamic spatial relationships and depth [Ray, 1935]. By bridging fine art photography from the 20th century with data visualization, RAY creates contemporary Rayograph that relate back to the past.

Portions of the visualization are then selected and shown in turn, determined by the regions of interest in the images that the computer vision system can interpret. The corresponding textual element (the words generated by DenseCap [Johnson et al., 2016]) is visualized in the regions with the moving pixels as an assemblage of pictorial fragments and semantic meanings. The system stores the visualization frames from the current to the past 29 frames and presents each frame in a transparent layer (30 layers in total)

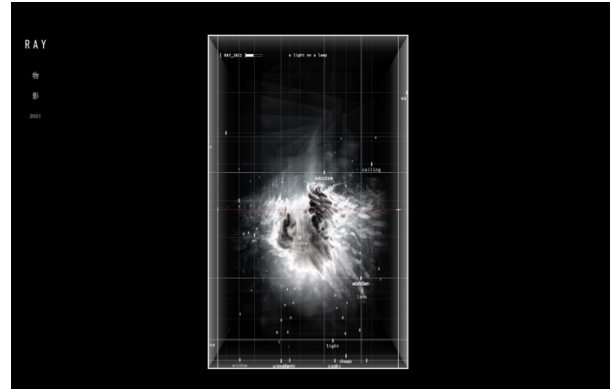


Figure 7: A visualization generated by RAY based on observation of a participant via a camera, 2022. (©Weidi Zhang.)

with different rotation angles to present the visualization as virtual sculptures that traces the passage of time.

Inspired by the Advanced Image Technology (AIT) used in the airport that reveals information of potential threats, the design of RAY's interface presents layers of information as a body scanner interface, including virtual sculpture (main visualization), textual elements (descriptive words), a 3D frame, and indicators.

4.3 Sound and installation design

In the installation, the ideal projection size of the visualization is approximately at human scale: 2 meters in height. A camera is hidden in the space to capture audience presence and send the live streaming feed to the system RAY, which results in an ever-changing interactive experience. The sound design uses field recordings of nature with transformation using the technique of modular synth. The audio is not interactive but has a looped sound composition. When the audiences enter the space and start reading the textual elements system represented, they usually realize that this is a visualization generated based on their presence. After the audience moves around the space, (approximately 5 minutes), they realize that their movements transform the visualization. The hidden camera captures the audience – like countless cameras in public spaces do – evaluate the information based on the intelligent system's prior knowledge and worldview, leading to an indeterminate visualization.

Since 2022, RAY has been exhibited at conferences, galleries, and festivals in Canada, the United States, Spain, the Netherlands, and Korea. It engages participants from diverse cultural backgrounds. Participants intuitively assumed the texts described the surroundings captured by the camera, so they moved, tried different gestures, and poses, and held some interesting objects in their pockets in front of the cameras (Figure 8). By viewing real-time language output on screen and interacting with it, people automatically build connections for meaning making. They usually perceive inaccurate descriptions of surroundings as an error by the machine, and the participants embraced this mistake by actively interacting to observe the uncertainty of the system and offsets of interpretations.

With its abstract visual language, RAY aims to build a universal language for communicating with participants. My conversations

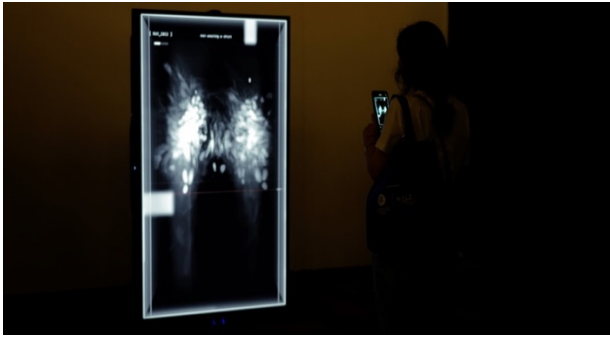


Figure 8: Installation View of *RAY* with participants, Vancouver Convention Centre, Canada, 2022. (©Weidi Zhang. Photo: Weidi Zhang.)

with the participants revealed that many of them were directly related to X-rays and surveillance. The scanning panel design and the monochrome aesthetic choice make them think of the moving images as their shadows generated by AI. Participants spend about five minutes in front of the screen, and some stay for a very short period of time if they do not feel the visuals provide instant feedback. There are some people who stay much longer than others because they are curious about the movements and concepts. Almost most of my participants choose to read the artist's statement after the interactions and started searching Rayograph immediately after hearing about the history of Rayograph.

Exhibiting this work internationally, *RAY* creates a synergy between abstraction and empathy. As people interact with the scanning screen, they feel a sense of insecurity and anxiety, and the abstract visualization and sound design contribute to the tension between humans and machines.

5 CONCLUSION

This paper presents the conceptual development, interactive intelligent system design, and experimental visualization strategies of the interactive AI art experience *RAY*. It engages audiences with an assemblage of virtual and actual as well as automation and originality through transforming live streaming image data into an

ever-evolving generative visualization in real time. *RAY* explores the aesthetics underlying the ideology of generative models and experiments with the rule-based interactive image-making process that re-contextualizes a cameraless photography technique (Rayograph). It aims to provide an art experience that criticizes the seeing of machines and implies the changing notion of images. *RAY* seeks poetic solutions and inspirations from 20th century fine art photography and re- interprets the artistic methods in the lens of AI and experimental visualization. *RAY* emphasizes the new aesthetic and culture reproduction that is co-created in an artist-machine-audience manner.

Acknowledgments

We acknowledge Prof. George Legrady and the Experimental Visualization Lab, as well as Prof. Marko Peljhan, for their invaluable support. This work was made possible through the SYMADES research grant from the Media Arts and Technology program at the University of California, Santa Barbara.

References

- MoMA Collection. 2019. Man Ray (Emmanuel Radnitzky) Rayograph 1923. Gallery Label from 2019. Retrieved January 16, 2022 from <http://www.moma.org/collection/works/44658>
- T. Paglen. 2019. Invisible images: Your pictures are looking at you. *Architectural Design*, 89(1), 22-27.
- I. Hoelzl, and R. Marie. 2015. *Softimage: Towards a New Theory of the Digital Image*. Intellect Books.
- H. Cohen. 1979. What is an image? In *Proceedings of the 6th International Joint Conference on Artificial Intelligence (IJCAI'79)*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 1028–1057.
- A. Mackenzie, and A. Munster. 2019. Platform seeing: Image ensembles and their invisibilities. *Theory, Culture & Society* 36, 5, 3-22.
- P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros. 2017. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1125-1134.
- M. Akten, R. Fiebrink, and M. Grierson. 2019. Learning to see: You are what you see. In *ACM SIGGRAPH 2019 Art Gallery (SIGGRAPH '19)*. Association for Computing Machinery, New York, NY, USA, Article 13, 1–6. DOI:<https://doi.org/10.1145/3306211.3320143>.
- J. Johnson, A. Karpathy, and L. Fei-Fei. 2016. Denscap: Fully convolutional localization networks for dense captioning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4565-4574.
- Stephen McNeill. 2020. *Alternative Photography. Photograms: A Magical Concept*. Retrieved January 16, 2022 from <https://www.alternativephotography.com/photograms-cameraless-photography>
- Man Ray. 1935. *Space Writing (Self-Portrait)*, 1935 by Man Ray. Retrieved January 19, 2023 from <https://www.manray.net/space-writing-self-portrait.jsp>